

The Very Basic Note in Algebra

Version 1.5.1

15-7-2023

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1 Functions

Definition 1.1. A *function*(函數) or a *mapping*(映射) is a relation between two sets A and B such that every element $a \in A$ corresponds to a unique element $b \in B$.

The set A is the *domain*(定義域) of the function while B is the *codomain*(陪域). We write $f : A \rightarrow B$ to denote a function f with domain A and codomain B . For each $a \in A$, we write $f(a)$ to denote the unique element in set B to which a corresponds under the mapping f . If $f(a) = b$, then b is called the *image*(像) of a , while a is called a *preimage*(原像) of b .

The set $\{f(a) : a \in A\}$ is called the *range*(值域) of the function, which means the set of all images under f . By definition, the range is a subset of B but may not necessarily be the same as B . Indeed, for each $b \in B$, it is possible that the equation $f(x) = b$ has no solution in $x \in A$, and it is possible that it has more than one solution.

Example 1.1. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = x^2 + 1$ for any $x \in \mathbb{R}$. Then both the domain and the codomain of f are $\mathbb{R}$. On the other hand, the range is the set $\{x^2 + 1 : x \in \mathbb{R}\} = \{y \in \mathbb{R} : y \geq 1\}$, which is a proper subset of the codomain.

Two functions are equal if and only if their domains, codomains, and the images of every element under the two functions are the same. In other words, even if two functions have the same ‘formula’, they may not be the same unless their domains and codomains are the same. If the domain and the codomain of a function are not specified, we usually assume implicitly that the domain is the largest set for which the formula is meaningful, while the codomain is taken to be the range.

Example 1.2. For the function with formula $f(x) = \sqrt{x-2} + 3$, we may assume its domain is $\{x \in \mathbb{R} : x \geq 2\}$ since the square root function is well-defined only on the set of nonnegative real numbers. Then the range of f is $\{y \in \mathbb{R} : y \geq 3\}$ since $\sqrt{x-2}$ can take any nonnegative value.

Definition 1.2. Let $f : A \rightarrow B$ be a function, and let C be a subset of A . The *restriction*(限制) of f on C is the function $f|_C : C \rightarrow B$ defined by $f|_C(x) = f(x)$ for any $x \in C$.

This means we change the domain of the function from A to C . Formally, the functions f and $f|_C$ are distinct, and so we should use different notations. However, sometimes we may still use f to mean both functions if there is no confusion.

Definition 1.3. Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be two functions. The *composition*(複合) of f and g is the function $g \circ f: A \rightarrow C$ defined by $(g \circ f)(x) = g(f(x))$ for any $x \in A$.

This means we first apply f and then g . The composition $g \circ f$ is only defined when the range of f is contained in the codomain of g . In particular, for $f: A \rightarrow A$, we can define $f \circ f$. This is denoted by $f^2(x)$. Similarly, we write $f^3(x)$ to mean $f(f(f(x)))$, etc. These are also known as *nested functions*(嵌套函數).

Example 1.3. Consider the functions $f(x) = 2x + 1$ and $g(x) = x + 2$. Both are defined on $\mathbb{R}$. We have

$$\begin{aligned}(g \circ f)(x) &= g(2x + 1) = 2x + 3, \\ (f \circ g)(x) &= f(x + 2) = 2x + 5.\end{aligned}$$

Note that $g \circ f \neq f \circ g$ in general even if both are well-defined.

Definition 1.4. Let A be a nonempty set. The *identity function*(恆等函數) on A is the function $f: A \rightarrow A$ defined by $f(x) = x$ for any $x \in A$.

Definition 1.5. A *constant function*(常數函數) is a function $f: A \rightarrow B$ such that the image $f(x)$ is the same for all $x \in A$.

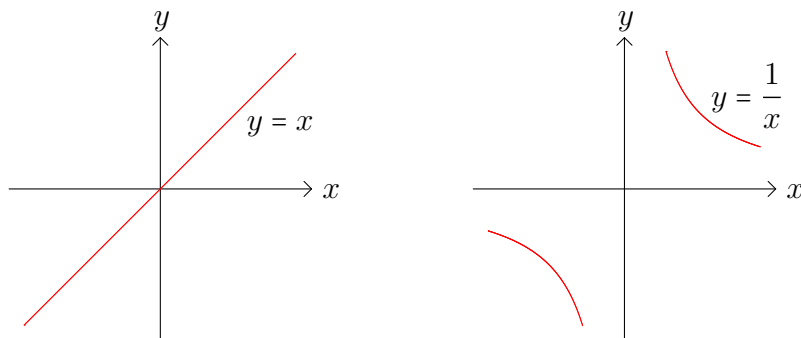
Definition 1.6. Let $f: A \rightarrow B$ be a function. The *inverse*(反函數) of f is the function $f^{-1}: B \rightarrow A$ such that $f^{-1} \circ f$ is the identity function on A , and $f \circ f^{-1}$ is the identity function on B .

Equivalently, this means $(f^{-1} \circ f)(x) = x$ for all $x \in A$, and $(f \circ f^{-1})(y) = y$ for all $y \in B$. Note that the inverse may not exist. However, it must be unique if it exists.

Definition 1.7. The *graph*(圖) of a function $f: A \rightarrow B$ is the set $\{(a, f(a)) : a \in A\}$.

The graph of a function $f: A \rightarrow B$ is a subset of $A \times B$. In particular, when $A = B = \mathbb{R}$, the graph is a subset of $\mathbb{R}^2$ so that we can represent each element in the graph as a point on the Cartesian xy -plane. It is sometimes convenient for us to study a real-valued function with its graph.

Example 1.4. The following show the graphs of $f(x) = x$ and $g(x) = \frac{1}{x}$ (for $x \neq 0$) respectively.



Definition 1.8. A function $f : A \rightarrow B$ is called *injective*(單射) or *one-to-one* if $f(a_1) = f(a_2)$ implies $a_1 = a_2$ for any $a_1, a_2 \in A$.

In other words, the images of distinct elements are distinct.

Definition 1.9. A function $f : A \rightarrow B$ is called *surjective*(滿射) or *onto* if for any $b \in B$, there exists $a \in A$ such that $f(a) = b$.

In other words, the range is the same as the codomain of the function.

Definition 1.10. A function $f : A \rightarrow B$ is called *bijective*(雙射) or *one-to-one correspondence* if it is both injective and surjective.

For a bijective function, the elements in the domain and the codomain can be paired up under the function. In particular, if both sets are finite, then they have the same number of elements.

Proposition 1.1. Let $f : A \rightarrow B$ be a function. Then f has an inverse if and only if f is bijective.

To find the inverse, we simply need to assign to each element $b \in B$ the unique element in A which maps to b under f . The existence of such an element is due to the surjectivity of f , while the uniqueness is guaranteed by the injectivity.

Example 1.5. We shall show that the function $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ defined by

$$f(x) = \frac{1}{x^2}$$

is bijective. Indeed, if $f(x) = f(y)$, then $\frac{1}{x^2} = \frac{1}{y^2}$ and hence $x^2 = y^2$. Since the domain of f is $\mathbb{R}^+$, both x and y are positive. Therefore, we must have $x = y$, implying that f is injective.

Next, for any $z \in \mathbb{R}^+$, we solve $f(w) = z$ to obtain $w = \frac{1}{\sqrt{z}}$, which belongs to $\mathbb{R}^+$. This shows f is surjective. Therefore, f is bijective. Indeed, the inverse of f is exactly the function $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ defined by

$$g(y) = \frac{1}{\sqrt{y}}.$$

Note that whether or not a function is injective and/or surjective depends on the domain and the codomain. In the above example, if we change the domain to $\mathbb{R} - \{0\}$, it is no longer injective. If we change the codomain to $\mathbb{R}$, it is no longer surjective.

Definition 1.11. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *bounded* (有界) if there exists a constant M such that $|f(x)| \leq M$ for any $x \in \mathbb{R}$. Otherwise, f is said to be *unbounded* (無界).

If there is a constant M such that $f(x) \leq M$ for any $x \in \mathbb{R}$, we say that f is *bounded above* (有上界). Similarly, f is *bounded below* (有下界) if there is a constant m such that $f(x) \geq m$ for any $x \in \mathbb{R}$. Note that f is bounded if and only if it is both bounded above and bounded below. For an unbounded function, it can be as large as possible, or as small as possible (or both).

Definition 1.12. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called *monotonic increasing* (單調遞增) if $f(x) \leq f(y)$ for any $x < y$. It is called *monotonic decreasing* (單調遞減) if $f(x) \geq f(y)$ for any $x < y$. In either case, it is called a *monotonic function* (單調函數).

In particular, f is called *strictly increasing* (嚴格遞增) if $f(x) < f(y)$ for any $x < y$. Similarly, it is called *strictly decreasing* (嚴格遞減) if $f(x) > f(y)$ for any $x < y$.

Note that a strictly increasing or decreasing function must be injective.

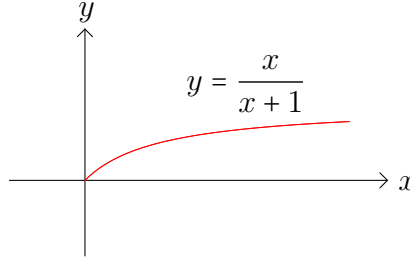
Example 1.6. We shall show that the function $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ defined by

$$f(x) = \frac{x}{x+1}$$

is strictly increasing. Indeed, for any $x < y$, we have

$$\frac{x}{x+1} - \frac{y}{y+1} = \frac{x-y}{(x+1)(y+1)} < 0$$

since x, y are positive. However, note that $f(x) < 1$ for any $x \in \mathbb{R}^+$. This shows that a strictly increasing function may not necessarily increase to infinity. The following figure shows the graph of f .



Definition 1.13. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called an *even function*(偶函数) if $f(x) = f(-x)$ for any $x \in \mathbb{R}$, and is called an *odd function*(奇函数) if $f(x) = -f(-x)$ for any $x \in \mathbb{R}$.

Proposition 1.2. We have the following facts about even functions and odd functions.

- (a) The sum of two even functions is an even function.
- (b) The sum of two odd functions is an odd function.
- (c) The product of two even functions is an even function.
- (d) The product of two odd functions is an even function.
- (e) The product of an even function and an odd function is an odd function.

Definition 1.14. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called a *periodic function*(周期函数) if there exists a constant $d > 0$ such that $f(x+d) = f(x)$ for any $x \in \mathbb{R}$. In that case, d is called a *period*(周期) of f .

Usually, we use the same word ‘period’ to mean the smallest period of a periodic function. However, the smallest period needs not exist. For a counterexample, please see section 1 of **Functions 5.0.1**.

Example 1.7. The trigonometric functions $\sin x$, $\cos x$ and $\tan x$ are well-known examples of periodic functions (we can still talk about the periodicity even though the domain of $\tan x$ is not $\mathbb{R}$). Their smallest periods are 2π , 2π and π respectively.

2 Polynomials

2.1 Basic Facts about Polynomials

Definition 2.1. A real *polynomial*(多項式) is a function $P: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

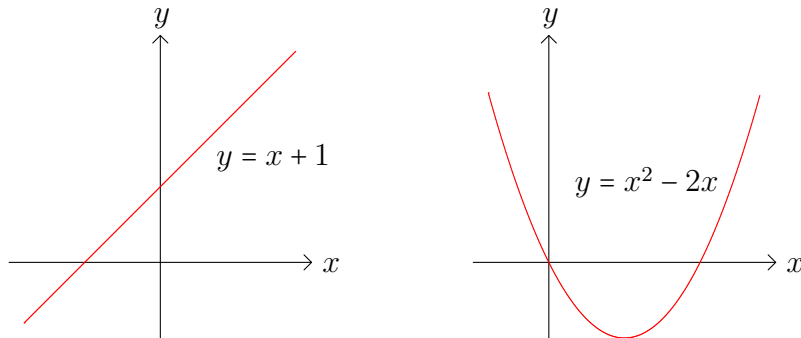
where $a_n, a_{n-1}, \dots, a_1, a_0$ are real constants. In that case, $a_n, a_{n-1}, \dots, a_1, a_0$ are called the *coefficients*(係數) of the polynomial $P(x)$. In particular, a_0 is the *constant term*(常數項).

We use the notation $\mathbb{R}[x]$ to denote the set of all real polynomials. Similarly, if $a_n, a_{n-1}, \dots, a_1, a_0$ are integers, we say that $P(x)$ is a polynomial with integer coefficients and write $P(x) \in \mathbb{Z}[x]$. The zero polynomial $P(x) = 0$ is a trivial example of a polynomial. A polynomial $P(x)$ is said to be a *monic polynomial*(首一多項式) if its leading coefficient a_n is equal to 1.

Definition 2.2. Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a nonzero polynomial with $a_n \neq 0$. Then n is called the *degree*(次數) of $P(x)$, and is denoted by $\deg(P(x))$ or $\deg P$.

By convention, the degree of the zero polynomial is defined to be negative infinity. When $\deg P = 0$ or P is the zero polynomial, i.e. $P(x) = c$ for some constant c , P is called a *constant polynomial*(常數多項式). When $\deg P = 1$ and $\deg P = 2$, P is called a *linear polynomial*(線性多項式) and a *quadratic polynomial*(二次多項式) respectively.

Example 2.1. By definition, $x+1$ is a linear polynomial, while x^2-2x is a quadratic polynomial. The following figures show the graphs of these polynomials. Note that the graph of a linear polynomial is a straight line, while that of a quadratic polynomial is a *parabola*(拋物線).



Proposition 2.1. Let $P(x)$ and $Q(x)$ be nonzero polynomials. Then the following hold.

- (a) $\deg(P + Q) \leq \max\{\deg P, \deg Q\}$
- (b) $\deg(PQ) = \deg P + \deg Q$

In particular, when $\deg P > \deg Q$, then $\deg(P + Q) = \deg P$.

Example 2.2. Let $P(x) = x^2 - x + 1$ and $Q(x) = x - 3$. Then $P(x) + Q(x) = x^2 - 2$ has degree $2 = \deg P$ and $P(x)Q(x) = x^3 - 4x^2 + 4x - 3$ has degree $3 = \deg P + \deg Q$.

Proposition 2.2. Let $P(x)$ and $Q(x)$ be two nonzero polynomials. Then the following hold.

- (a) $P(x)$ is unbounded
- (b) if $\deg P > \deg Q$, then there exists a constant M such that $|P(x)| > |Q(x)|$ for any x satisfying $|x| > M$

Although a formal proof can be given, the results are quite intuitive. They tell us the growth of polynomials. As a remark, all polynomials grow less rapidly than exponential functions $f(x) = a^x$ where $a > 1$. On the other hand, all non-constant polynomials grow more rapidly than the logarithm function $f(x) = \log x$ or $\ln x$.

Theorem 2.1. (Division algorithm(除法算則)) Let $P_1(x), P_2(x) \in \mathbb{R}[x]$ be polynomials where P_2 is nonzero. Then there exist unique $Q(x), R(x) \in \mathbb{R}[x]$ such that

$$P_1(x) = Q(x)P_2(x) + R(x) \quad \text{and} \quad \deg R < \deg P_2.$$

The idea of the proof is to apply long division until we obtain a polynomial of degree less than that of P_2 . Analogous to the case of integers, $Q(x)$ is called the *quotient*(商) while $R(x)$ is called the *remainder*(餘數). When R is the zero polynomial, P_2 is called a *factor*(因子) of P_1 .

Example 2.3. Let $P_1(x) = x^2 - 4x + 3$ and $P_2(x) = x + 2$. By simple calculation, we have $P_1(x) = P_2(x)(x - 6) + 15$, where $\deg(15) = 0 < 1 = \deg(x + 2)$. In addition, since $P_1(x) = (x - 1)(x - 3)$, both $x - 1$ and $x - 3$ are factors of P_1 .

Theorem 2.2. (Remainder theorem(餘數定理)) For any polynomial $P(x)$, the remainder of P when divided by a linear polynomial $x - a$ is $P(a)$.

Proof. By the division algorithm, we have $P(x) = Q(x)(x - a) + R(x)$ for some $Q(x), R(x) \in \mathbb{R}[x]$ such that $\deg R < \deg(x - a)$. As $\deg(x - a) = 1$, R must be a constant polynomial. Let $R(x) = r$. Then we have

$$P(x) = Q(x)(x - a) + r.$$

Putting $x = a$, we get $P(a) = r$, i.e. the remainder is simply $P(a)$. □

Theorem 2.3. (Factor theorem(因式定理)) For any polynomial $P(x)$, the linear polynomial $x - a$ is a factor of P if and only if $P(a) = 0$.

This is a simple consequence of the remainder theorem.

Proposition 2.3. A polynomial is an even function if and only if the coefficients of every odd power is zero. Similarly, a polynomial is an odd function if and only if the coefficients of every even power is zero.

Example 2.4. For example, the polynomial $P(x) = 2x^4 + 4x^2 - 3$ is an even function while $Q(x) = x^3 - 3x$ is an odd function.

2.2 Polynomial Equations

Definition 2.3. A *polynomial equation*(多项式方程) is an equation $P(x) = 0$ where P is a polynomial.

In particular, when $\deg P = 1$ and $\deg P = 2$, the equation $P(x) = 0$ is called a *linear equation*(线性方程) and a *quadratic equation*(二次方程) respectively.

Definition 2.4. A real number a is called a *solution*(解) or a *root*(根) of a polynomial equation $P(x) = 0$ if $P(a) = 0$.

Sometimes we simply say a is a root of $P(x)$.

Proposition 2.4. A real number a is a root of $P(x) = 0$ if and only if $x - a$ is a factor of $P(x)$.

This is a consequence of the factor theorem. If $(x - a)^k$ is a factor of $P(x)$ for some $k \geq 1$ while $(x - a)^{k+1}$ is not, we say that a is a root of *multiplicity* (重數) k of $P(x)$. For $k = 1$, a is called a *simple root* (單根). For $k \geq 2$, a is called a *multiple root* (重根).

Example 2.5. Since $(3)^3 - 2(3)^2 - 4(3) + 3 = 0$, 3 is a root of $x^3 - 2x^2 - 4x + 3 = 0$. By *proposition 2.4*, $x - 3$ is a factor of $x^3 - 2x^2 - 4x + 3$. After factorization, we can rewrite the equation as $(x - 3)(x^2 + x - 1) = 0$. Since $(3)^2 + (3) - 1 = 11 \neq 0$, $x - 3$ is not a factor of $x^2 + x - 1$. So 3 is a simple root of the original equation.

Proposition 2.5. The polynomial equation $P(x) = 0$ with $\deg P = n \geq 0$ has at most n real roots (counted with multiplicity).

Proof. Suppose on the contrary that it has at least $n + 1$ roots $a_1, a_2, \dots, a_{n+1}$. By repeatedly using *proposition 2.4*, we can factor out the linear factors so that

$$P(x) = (x - a_1)(x - a_2) \cdots (x - a_{n+1})Q(x)$$

for some nonzero polynomial Q . By a generalization of *proposition 2.1(b)*, we have

$$\deg P = \deg(x - a_1) + \deg(x - a_2) + \cdots + \deg(x - a_{n+1}) + \deg Q \geq n + 1,$$

which is impossible as $\deg P = n$. So $P(x) = 0$ has at most n real roots. $\square$

Proposition 2.6. The only real root to the linear equation $ax + b = 0$ is $-\frac{b}{a}$.

This is obvious. Note that the linear condition guarantees that $a \neq 0$.

Definition 2.5. Let $P(x) = ax^2 + bx + c$ be a quadratic polynomial where a, b, c are real constants with $a \neq 0$. The *discriminant* (判別式) of $P(x)$, denoted by Δ , is defined by $\Delta = b^2 - 4ac$.

Proposition 2.7. Let $ax^2 + bx + c = 0$ be a quadratic equation and let $\Delta = b^2 - 4ac$ be its discriminant.

(a) If $\Delta \geq 0$, then the equation has two real roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

In particular, when $\Delta = 0$, the equation has a multiple root. When $\Delta > 0$, the equation has two distinct real roots.

(b) If $\Delta < 0$, then the equation has no real roots.

Proof. The equation $ax^2 + bx + c = 0$ can be rewritten as $4a^2x^2 + 4abx + 4ac = 0$. This is the same as

$$(2ax + b)^2 = b^2 - 4ac.$$

It is now obvious that the equation is solvable only if $\Delta \geq 0$. In that case, we can take square roots from both sides to obtain $2ax + b = \pm\sqrt{b^2 - 4ac}$, i.e.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

□

There are also formulae and standard methods to find all the roots of a *cubic* (三次) (degree 3) or *quartic* (四次) (degree 4) polynomial equations. But the methods are clumsy and there are in general no formulae to solve polynomial equations of degree at least 5. Therefore, a better way to find a root of an equation is by inspection or trial and error.

Theorem 2.4. (*Rational root theorem* (有理根定理)) Consider a nonzero polynomial $P(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ with integer coefficients. Let $\frac{u}{v}$ be a rational root to the equation $P(x) = 0$ where u, v are relatively prime integers. Then $v \mid a_n$ and $u \mid a_0$.

Proof. By definition, we have

$$a_n \left(\frac{u}{v}\right)^n + a_{n-1} \left(\frac{u}{v}\right)^{n-1} + \cdots + a_1 \left(\frac{u}{v}\right) + a_0 = 0$$

so that

$$a_n u^n + a_{n-1} u^{n-1} v + \cdots + a_1 u v^{n-1} + a_0 v^n = 0.$$

This implies $v \mid a_n u^n$. As $\gcd(u, v) = 1$, we obtain $v \mid a_n$. Similarly, we get $u \mid a_0 v^n$, and hence $u \mid a_0$. □

The rational root theorem gives us a useful way to test for the rational roots of a polynomial equation with integer coefficients, which is sufficient for our use in most cases.

Example 2.6. Consider the equation $6x^3 + 11x^2 - 3x - 2 = 0$. If it has a rational root $\frac{u}{v}$, then $v \mid 6$ and $u \mid -2$ by the rational root theorem. Also, we need $\gcd(u, v) = 1$. Without loss of generality, we may further assume $u > 0$. Then the only possible roots are

$$\pm 1, \pm \frac{1}{2}, \pm \frac{1}{3}, \pm \frac{1}{6}, \pm 2, \pm \frac{2}{3}.$$

By testing, we notice that $\frac{1}{2}, -\frac{1}{3}$ and -2 are roots to the equation. This gives

$$6x^3 + 11x^2 - 3x - 2 = (2x - 1)(3x + 1)(x + 2).$$

Example 2.7. We sometimes come across polynomials with coefficients in square roots. Consider the equation

$$2x^3 - \sqrt{3}x^2 - x - 2\sqrt{3} = 0.$$

Since only the even powers involve the term $\sqrt{3}$, it is reasonable to consider the substitution $x = \sqrt{3}y$. Then the equation becomes $6y^3 - 3y^2 - y - 2 = 0$. Using the same method as above or by inspection, we find that $y = 1$ is a solution. So the original equation has a factor $x - \sqrt{3}$. Hence, we get

$$(x - \sqrt{3})(2x^2 + \sqrt{3}x + 2) = 0,$$

which shows that $x = \sqrt{3}$ is the only real solution as the discriminant of $2x^2 + \sqrt{3}x + 2$ is $-13 < 0$.

In general, even though we may not find any real roots of an equation explicitly, we can still obtain certain information of the roots from the equation.

Theorem 2.5. (**Vieta's theorem**(韋達定理)) Let $n \geq 1$. Consider the polynomial equation

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$$

with n real roots $b_1, b_2, \dots, b_n$ (counted with multiplicity). Then

$$\begin{aligned} b_1 + b_2 + \cdots + b_n &= -\frac{a_{n-1}}{a_n}, \\ b_1 b_2 + b_1 b_3 + \cdots + b_{n-1} b_n &= \frac{a_{n-2}}{a_n}, \\ &\vdots \\ b_1 b_2 \cdots b_n &= (-1)^n \frac{a_0}{a_n}. \end{aligned}$$

In general, the sum of all products of k roots is equal to $(-1)^k \frac{a_{n-k}}{a_n}$.

Proof. Since each b_j is a root, $x - b_j$ is a factor of the polynomial. By comparing the leading coefficients, we find that

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = a_n (x - b_1)(x - b_2) \cdots (x - b_n).$$

Expanding the right-hand side and comparing the coefficients, we obtain the desired result. $\square$

Example 2.8. If the quadratic equation $ax^2 + bx + c = 0$ has two real roots α, β , then

$$\alpha + \beta = -\frac{b}{a},$$

$$\alpha\beta = \frac{c}{a}.$$

This can also be verified directly by solving α and β explicitly.

Example 2.9. If the cubic equation $ax^3 + bx^2 + cx + d = 0$ has three real roots α, β, γ , then

$$\alpha + \beta + \gamma = -\frac{b}{a},$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a},$$

$$\alpha\beta\gamma = -\frac{d}{a}.$$

2.3 Identities

In the following, we list out some useful identities involving polynomials of more variables.

Proposition 2.8. Let a, b, c be real numbers, and let n be a positive integer. Then the following hold.

- (a) $(a + b)^2 = a^2 + 2ab + b^2$
- (b) $(a - b)^2 = a^2 - 2ab + b^2$
- (c) $a^2 - b^2 = (a - b)(a + b)$
- (d) $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- (e) $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- (f) $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$
- (g) $a^n + b^n = (a + b)(a^{n-1} - a^{n-2}b + \dots + b^{n-1})$ if n is odd
- (h) $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
- (i) $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}((a - b)^2 + (b - c)^2 + (c - a)^2)$
- (j) $(a^2 + b^2)(c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2$

3 Complex Numbers

Definition 3.1. The number i is defined to be the *imaginary number*(虛數) satisfying $i^2 = -1$.

Note that there are two imaginary numbers satisfying $x^2 = -1$, namely, i and $-i$. Due to symmetry, we may choose one of these solutions and denote it by i . It is customary to write $\sqrt{-1}$ to represent the symbol i . However, this notation should not be mixed up with the usual square root. The usual square root sign is a function from $\mathbb{R}_0^+$ to $\mathbb{R}_0^+$. The term $\sqrt{-1}$ is undefined under this definition since -1 does not belong to its domain. A better understanding is to regard $\sqrt{-1}$ as a single symbol, namely, i . By convention, we also write $\sqrt{-a}$ to mean $\sqrt{a}i$ for any real number $a \geq 0$.

Definition 3.2. The set of all numbers of the form $a + bi$ where $a, b \in \mathbb{R}$ is called the set of *complex numbers*(複數), and is denoted by $\mathbb{C}$.

Let $z = a + bi$ be a complex number. When $b = 0$, $z = a \in \mathbb{R}$ can be any real number. So each real number is in fact a complex number. In general, we call a the *real part*(實部) of z , denoted by $\text{Re}(z)$, and call b the *imaginary part*(虛部) of z , denoted by $\text{Im}(z)$. When $a = 0$, $z = bi$ is called a *purely imaginary number*(純虛數) (some may require $b \neq 0$). We use $i\mathbb{R}$ to denote the set of all purely imaginary numbers.

Two complex numbers are equal if and only if both their real parts and imaginary parts are the same. We can apply algebraic operations on complex numbers like what we do on real numbers. We may regard i as a variable and apply the fact $i^2 = -1$ whenever needed. Indeed, for any complex numbers $w = a + bi$ and $z = c + di$, we have

$$\begin{aligned} w + z &= (a + bi) + (c + di) = (a + c) + (b + d)i, \\ w - z &= (a + bi) - (c + di) = (a - c) + (b - d)i, \\ wz &= (a + bi)(c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad + bc)i, \end{aligned}$$

and for $z \neq 0$ (i.e. $(c, d) \neq (0, 0)$),

$$\frac{w}{z} = \frac{a + bi}{c + di} = \frac{a + bi}{c + di} \cdot \frac{c - di}{c - di} = \frac{ac - adi + bci - bdi^2}{c^2 - (di)^2} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i.$$

Definition 3.3. Let $z = a + bi$ be a complex number. Then its *complex conjugate*(共軛複數) is defined to be $a - bi$, and is denoted by $\bar{z}$.

Definition 3.4. Let $z = a + bi$ be a complex number. Then its *modulus*(模) is defined to be $\sqrt{a^2 + b^2}$, and is denoted by $|z|$.

Note that the modulus of a complex number must be a nonnegative real number. In particular, when z is a real number, then its modulus is simply its absolute value.

Example 3.1. Let $z = 2 + 3i$. Then $\bar{z} = 2 - 3i$ and $|z| = \sqrt{2^2 + 3^2} = \sqrt{13}$.

Proposition 3.1. Let z and w be complex numbers, and let a be a real number. Then the following hold.

- (a) $\overline{\bar{a}} = a$
- (b) $\overline{\bar{z}} = z$
- (c) $\overline{az} = a\bar{z}$
- (d) $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$
- (e) $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$
- (f) $\overline{z \pm w} = \bar{z} \pm \bar{w}$
- (g) $\overline{zw} = \bar{z} \times \bar{w}$
- (h) $\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}$ if $w \neq 0$
- (i) $|zw| = |z||w|$
- (j) $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$ if $w \neq 0$
- (k) ([Triangle inequality](#) (三角不等式)) $|z + w| \leq |z| + |w|$
- (l) $|\bar{z}| = |z|$
- (m) $|z|^2 = z\bar{z}$

Example 3.2. Let $z = 2 + 3i$ and $w = 1 - i$. Then $zw = (2 + 3i)(1 - i) = 5 + i$. This implies

$$\begin{aligned}\bar{z} \times \bar{w} &= (2 - 3i)(1 + i) = 5 - i = \overline{zw}, \\ |zw| &= |5 + i| = \sqrt{26} = \sqrt{13}\sqrt{2} = |z||w|.\end{aligned}$$

These verify [proposition 3.1\(g\)](#) and [\(i\)](#) in this case.

Next, we check that

$$|z + w| = |3 + 2i| = \sqrt{13} < \sqrt{13} + \sqrt{2} = |z| + |w|,$$

so that the triangle inequality is satisfied.

In addition, we have

$$\begin{aligned} |\bar{z}| &= |2 - 3i| = \sqrt{13} = |z|, \\ |z|^2 &= 13 = (2 + 3i)(2 - 3i) = z\bar{z}. \end{aligned}$$

These agree with [proposition 3.1\(l\)](#) and (m).

Analogous to the case of real numbers, we can also talk about polynomials with complex coefficients. All definitions and results in [section 2](#) apply to complex polynomials. In the following, we go through some additional results.

Proposition 3.2. If z is a complex root to a real polynomial equation $P(x) = 0$, then $\bar{z}$ is also a root to this equation.

Proof. Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$. It is given that

$$a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0 = 0.$$

By taking conjugates on both sides, we obtain

$$\overline{a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0} = 0,$$

and hence $a_n \bar{z}^n + a_{n-1} \bar{z}^{n-1} + \cdots + a_1 \bar{z} + a_0 = 0$ with the use of [proposition 3.1\(c\)](#) and (f) (note that $a_n, a_{n-1}, \dots, a_1, a_0 \in \mathbb{R}$). By definition, $\bar{z}$ is a root to the equation. $\square$

Theorem 3.1. ([Fundamental theorem of algebra](#)(代數基本定理)) All non-constant complex polynomials have a complex root.

The proof of this theorem is beyond our discussion. As a corollary, by factoring each linear factor one by one, we see that every non-constant complex polynomial can be completely factorized into linear factors. Note that this implies a complex polynomial equation of degree n has exactly n complex roots (counted with multiplicity). One may compare this result with [proposition 2.5](#).

Proposition 3.3. All non-constant real polynomials can be completely factorized into linear and quadratic factors (with real coefficients).

Proof. Let $P(x)$ be a non-constant real polynomial. By the fundamental theorem of algebra, there is a root z for $P(x)$. If z is real, then $x - z$ is a linear factor of $P(x)$ by the factor theorem. If z is not real, then [proposition 3.2](#) suggests $\bar{z}$ and z are distinct roots of $P(x)$. Therefore,

$$(x - z)(x - \bar{z}) = x^2 - (z + \bar{z})x + z\bar{z}$$

is a quadratic factor of $P(x)$, where $z+\bar{z} = 2\operatorname{Re}(z)$ and $z\bar{z} = |z|^2$ are both real numbers. In other words, $P(x)$ has a linear or a quadratic factor with real coefficients. By taking out this factor, we can carry out the same procedure to obtain more linear or quadratic factors of $P(x)$, until we get a constant polynomial. Therefore, $P(x)$ is a product of linear and quadratic factors with real coefficients. $\square$

4 Sequences

Definition 4.1. A real *sequence*(數列) is a function with domain $\mathbb{N}$ or $\mathbb{N}_0$ and taking real values. We usually denote the images as a_n instead of $a(n)$. A sequence is usually expressed in the form $\{a_n\}_{n \geq 1}$ or $(a_n)_{n \geq 1}$.

Since a sequence is a function, many definitions introduced in section 1 apply to sequences. For example, a sequence $\{a_n\}$ is *bounded*(有界) if there exists a constant M such that $|a_n| \leq M$ for any $n \in \mathbb{N}$. It is *monotonic*(單調) if $a_1 \leq a_2 \leq \dots$ or $a_1 \geq a_2 \geq \dots$.

Definition 4.2. A *recurrence sequence*(遞迴數列) is a sequence for which every term except some initial terms is defined by its preceding terms.

The equation which expresses a_n in terms of its preceding terms is called a *recurrence relation*(遞迴關係式). A recurrence relation is called *linear*(線性) if

$$a_n = f_1(n)a_{n-1} + f_2(n)a_{n-2} + \dots + f_k(n)a_{n-k} + g(n)$$

for some integer $k \geq 1$ and some functions $f_1, f_2, \dots, f_k, g$. When $f_1, f_2, \dots, f_k$ are constant functions, the recurrence relation is said to have constant coefficients. When g is the zero function, the relation is said to be *homogeneous*(齊次). If f_k is not the zero function, then the relation has *order*(階) k .

Example 4.1. The sequence $\{a_n\}$ defined by $a_n = 3a_{n-1} - 2a_{n-2} + 1$ is a second order linear recurrence relation with constant coefficients. However, it is not homogeneous due to the additional term 1.

The sequence $\{b_n\}$ defined by $b_n = n^2b_{n-1} + b_{n-2}$ is also linear by definition, even though n^2 is a quadratic polynomial. It is homogeneous.

Definition 4.3. An *arithmetic sequence*(等差數列) is a sequence $\{a_n\}$ with recurrence relation $a_{n+1} = a_n + d$ for any $n \geq 1$ where d is a constant.

This means the difference between any two consecutive terms is a constant. We call d the *common difference*(公差) of the arithmetic sequence.

Proposition 4.1. Let $\{a_n\}$ be an arithmetic sequence with $a_{n+1} = a_n + d$ for all $n \geq 1$. Then $a_n = a_1 + (n-1)d$ for all $n \geq 1$.

Proof. We have $a_n = a_{n-1} + d = a_{n-2} + 2d = \dots = a_1 + (n-1)d$. □

Proposition 4.2. Let $\{a_n\}$ be an arithmetic sequence with $a_{n+1} = a_n + d$ for all $n \geq 1$. Then

$$a_1 + a_2 + \cdots + a_n = \frac{n(a_1 + a_n)}{2} = \frac{n(2a_1 + (n-1)d)}{2}.$$

Proof. Let $S = a_1 + a_2 + \cdots + a_n$. Note that

$$2S = (a_1 + a_n) + (a_2 + a_{n-1}) + \cdots + (a_n + a_1).$$

For each k , we have

$$a_k + a_{n+1-k} = (a_1 + (k-1)d) + (a_1 + (n-k)d) = 2a_1 + (n-1)d.$$

This proves $S = \frac{n(2a_1 + (n-1)d)}{2}$, which is also equal to $\frac{n(a_1 + a_n)}{2}$. $\square$

Definition 4.4. A *geometric sequence* (等比數列) is a sequence $\{a_n\}$ with recurrence relation $a_{n+1} = ra_n$ for any $n \geq 1$ where r is a constant.

This means the ratio between any two consecutive terms is a constant. We call r the *common ratio* (公比) of the geometric sequence.

Proposition 4.3. Let $\{a_n\}$ be a geometric sequence with $a_{n+1} = ra_n$ for all $n \geq 1$. Then $a_n = r^{n-1}a_1$ for all $n \geq 1$.

Proof. We have $a_n = ra_{n-1} = r^2a_{n-2} = \cdots = r^{n-1}a_1$. $\square$

Proposition 4.4. Let $\{a_n\}$ be a geometric sequence with $a_{n+1} = ra_n$ for all $n \geq 1$ where $r \neq 1$. Then

$$a_1 + a_2 + \cdots + a_n = \frac{a_1(r^n - 1)}{r - 1}.$$

Proof. Let

$$S = a_1 + a_2 + \cdots + a_n = a_1(1 + r + r^2 + \cdots + r^{n-1}).$$

Multiplying both sides by r , we obtain

$$rS = a_1(r + r^2 + r^3 + \cdots + r^n).$$

It follows that

$$(r-1)S = rS - S = a_1(r^n - 1).$$

The result follows readily. $\square$

If $|r| < 1$, we can compute the infinite sum $a + ar + ar^2 + \cdots = \frac{a}{1-r}$.

Example 4.2. The sequence $1, 3, 5, 7, \dots$ is arithmetic while $1, 2, 4, 8, \dots$ is geometric. Their general terms are $2n-1$ and 2^{n-1} respectively. Note that the recurrence relations for both arithmetic sequences and geometric sequences are linear and of first order. Also, we have

$$1 + 3 + 5 + \dots + (2n-1) = \frac{(1 + (2n-1))n}{2} = n^2,$$

$$1 + 2 + 4 + \dots + 2^{n-1} = \frac{2^n - 1}{2 - 1} = 2^n - 1.$$

Proposition 4.5. Let $\{a_n\}$ be a sequence with recurrence relation $a_{n+1} = ra_n + d$ for all $n \geq 1$. Then the general term is

$$a_n = r^{n-1}a_1 + (1 + r + r^2 + \dots + r^{n-2})d.$$

Proof. We simply need to sum up

$$\begin{aligned} a_n &= ra_{n-1} + d, \\ ra_{n-1} &= r^2a_{n-2} + rd, \\ r^2a_{n-2} &= r^3a_{n-3} + r^2d, \\ &\dots \\ r^{n-2}a_2 &= r^{n-1}a_1 + r^{n-2}d. \end{aligned}$$

□

Proposition 4.6. Let $\{a_n\}$ be a sequence with recurrence relation $a_{n+2} = ta_{n+1} + ua_n$ for all $n \geq 1$ where $u \neq 0$. Consider the *characteristic equation* (特徴方程)

$$\lambda^2 - t\lambda - u = 0.$$

(a) If it has two distinct (complex) roots α, β , then there exist constants A, B such that the general term is

$$a_n = A\alpha^n + B\beta^n. \quad (1)$$

(b) If it has a double root α , then there exist constants A, B such that the general term is

$$a_n = A\alpha^n + Bn\alpha^{n-1}. \quad (2)$$

Proof.

(a) Suppose α, β are two distinct roots of $\lambda^2 - t\lambda - u = 0$. Consider the equations

$$\begin{cases} A\alpha^2 + B\beta^2 = a_2, \\ A\alpha + B\beta = a_1. \end{cases}$$

Since α, β are distinct, there exists a unique solution

$$A = \frac{a_2 - a_1\beta}{\alpha(\alpha - \beta)}, \quad B = \frac{a_2 - a_1\alpha}{\beta(\beta - \alpha)}.$$

Suppose (1) is true for $n = k$ and $n = k + 1$. Then we have

$$\begin{aligned} a_{k+2} &= ta_{k+1} + ua_k \\ &= t(A\alpha^{k+1} + B\beta^{k+1}) + u(A\alpha^k + B\beta^k) \\ &= (t\alpha + u)A\alpha^k + (t\beta + u)B\beta^k \\ &= \alpha^2 \cdot A\alpha^k + \beta^2 \cdot B\beta^k \\ &= A\alpha^{k+2} + B\beta^{k+2}. \end{aligned}$$

Thus, (1) is true for all natural numbers n by induction (where the base cases hold by the choices of A and B).

(b) Similarly, if α is a double root of $\lambda^2 - t\lambda - u = 0$, i.e. $\alpha = \frac{t}{2}$, consider the equations

$$\begin{cases} A\alpha^2 + 2B\alpha^2 = a_2, \\ A\alpha + B\alpha = a_1. \end{cases}$$

Note that $\alpha \neq 0$ as $u \neq 0$. Therefore, there exists a unique solution

$$A = \frac{2a_1\alpha - a_2}{\alpha^2}, \quad B = \frac{a_2 - a_1\alpha}{\alpha^2}.$$

Suppose (2) is true for $n = k$ and $n = k + 1$. Then we have

$$\begin{aligned} a_{k+2} &= ta_{k+1} + ua_k \\ &= t(A\alpha^{k+1} + B(k+1)\alpha^{k+1}) + u(A\alpha^k + Bk\alpha^k) \\ &= (t\alpha + u)A\alpha^k + (t\alpha + u)Bk\alpha^k + tB\alpha^{k+1} \\ &= \alpha^2 \cdot A\alpha^k + \alpha^2 \cdot Bk\alpha^k + 2\alpha \cdot B\alpha^{k+1} \\ &= A\alpha^{k+2} + B(k+2)\alpha^{k+2}. \end{aligned}$$

Thus, (2) is true for all natural numbers n by induction.

□

Note that the constants A and B can be determined by the initial conditions, i.e. the values of a_1 and a_2 . Sometimes it may be more convenient to introduce the term a_0 (so that $a_2 = ta_1 + ua_0$) to solve the simultaneous equation.

Example 4.3. Suppose $\{a_n\}$ satisfies $a_{n+2} = 5a_{n+1} - 6a_n$ and $a_2 = 4$, $a_1 = 2$. To find its general term, we consider its characteristic equation $\lambda^2 - 5\lambda + 6 = 0$. It has two distinct roots $\lambda = 2, 3$. Therefore, by [proposition 4.6](#), there exist constants A, B such that $a_n = A \cdot 2^n + B \cdot 3^n$. Solving

$$\begin{cases} 4A + 9B = a_2 = 4, \\ 2A + 3B = a_1 = 2, \end{cases}$$

we obtain $A = 1$ and $B = 0$. Therefore, $a_n = 2^n$ for any $n \in \mathbb{N}$. This can also be proved by induction if the formula is correctly guessed.

5 Inequalities

Proposition 5.1. Let a, b, c, d be any real numbers. Then the following hold.

- (a) If $a \geq b$ and $b \geq c$, then $a \geq c$.
- (b) If $a \geq b$ and $c \geq d$, then $a + c \geq b + d$. In particular, $a \pm c \geq b \pm c$.
- (c) If $a \geq b$ and $c \geq 0$, then $ac \geq bc$.
- (d) If $a \geq b$ and $c \leq 0$, then $ac \leq bc$.
- (e) If $a \geq b \geq 0$ and $c \geq d \geq 0$, then $ac \geq bd$. In particular, $ab \geq 0$ and $a^2 \geq b^2$.
- (f) We have $a^2 \geq b^2$ if and only if $|a| \geq b \geq -|a|$.

Proposition 5.2. Let a be any real number. Then $a^2 \geq 0$, with equality $a = 0$.

For the word ‘equality’, we means that the two sides of the inequality $a^2 \geq 0$ are equal if and only if $a = 0$.

Proposition 5.3. Let $P(x) = ax^2 + bx + c$ be a quadratic polynomial where a, b, c are real constants with $a > 0$. Then the following hold.

- (a) If $\Delta \leq 0$, then $P(x) \geq 0$ for any real number x . Equality holds if and only if $\Delta = 0$ and $x = -\frac{b}{2a}$.
- (b) If $\Delta > 0$, then $P(x) \geq 0$ if and only if $x \geq \frac{-b + \sqrt{\Delta}}{2a}$ or $x \leq \frac{-b - \sqrt{\Delta}}{2a}$, while $P(x) \leq 0$ if and only if $\frac{-b - \sqrt{\Delta}}{2a} \leq x \leq \frac{-b + \sqrt{\Delta}}{2a}$.

Proof. See [proposition 1.4](#) of **Inequalities 5.0**. □

Theorem 5.1. ([AM-GM inequality](#)(均值不等式)) Let $n \geq 2$. For any positive real numbers $a_1, a_2, \dots, a_n$, we have

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}.$$

Equality holds if and only if $a_1 = a_2 = \dots = a_n$.

Proof. See [theorem 2.1](#) of **Inequalities 5.0**. □

Example 5.1. Let a, b, c be positive real numbers. By the AM-GM inequality, we have

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3 \sqrt[3]{\frac{a}{b} \cdot \frac{b}{c} \cdot \frac{c}{a}} = 3.$$

Equality holds when $\frac{a}{b} = \frac{b}{c} = \frac{c}{a}$, i.e. $a = b = c$. This shows the minimum value of $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is 3.

Theorem 5.2. (Cauchy-Schwarz inequality(柯西不等式)) Let $n \geq 2$. For any real numbers $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$, we have

$$(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \geq (a_1b_1 + a_2b_2 + \dots + a_nb_n)^2.$$

Equality holds if and only if $a_ib_j = a_jb_i$ for any $i \neq j$.

Proof. See theorem 2.5 of Inequalities 5.0. □

Note that the equality case can be interpreted as $a_1 : a_2 : \dots : a_n = b_1 : b_2 : \dots : b_n$. But in this notation, we use the convention that $a_j = 0$ if and only if $b_j = 0$ for any $1 \leq j \leq n$.

Example 5.2. Let a, b, c be positive real numbers satisfying $a + b + c = 6$. By the Cauchy-Schwarz inequality, we have

$$(a + b + c) \left(\frac{1}{a} + \frac{4}{b} + \frac{9}{c} \right) \geq \left(\sqrt{a} \cdot \sqrt{\frac{1}{a}} + \sqrt{b} \cdot \sqrt{\frac{4}{b}} + \sqrt{c} \cdot \sqrt{\frac{9}{c}} \right)^2 = 36.$$

This implies $\frac{1}{a} + \frac{4}{b} + \frac{9}{c} \geq 6$. Equality holds when

$$a : b : c = \frac{1}{a} : \frac{4}{b} : \frac{9}{c},$$

i.e. $a : b : c = 1 : 2 : 3$. Together with $a + b + c = 6$, this holds when $a = 1, b = 2, c = 3$.

This shows the minimum value of $\frac{1}{a} + \frac{4}{b} + \frac{9}{c}$ is 6.

Links

Theorems

- [AM-GM inequality](#)
- [Cauchy-Schwarz inequality](#)
- [Division algorithm \(polynomials\)](#)
- [Factor theorem](#)
- [Fundamental theorem of algebra](#)
- [Rational root theorem](#)
- [Remainder theorem](#)
- [Triangle inequality \(complex numbers\)](#)
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Propositions

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- 4.6: General term of second order linear homogeneous recurrence sequences with constant coefficients
- 5.1: Basic facts about inequalities
- 5.2: Nonnegativity of the square of real numbers
- 5.3: Quadratic inequalities

Terminologies and Notations

- $\mathbb{R}[x]$
- $\mathbb{Z}[x]$
- $\mathbb{C}$
- $i\mathbb{R}$
- arithmetic sequence
- bijective
- bounded above
- bounded below
- bounded function
- bounded sequence
- characteristic equation
- codomain
- coefficient
- common difference
- common ratio
- complex conjugate $\bar{z}$
- complex number
- composition $g \circ f$
- constant function
- constant polynomial
- constant term
- cubic polynomial
- degree $\deg P$
- discriminant Δ
- domain
- even function

- factor
- function $f : A \rightarrow B$
- geometric sequence
- graph
- homogeneous recurrence relation
- identity function
- image $f(a)$
- imaginary number i
- imaginary part $\text{Im}(z)$
- injective
- inverse function f^{-1}
- linear equation
- linear polynomial
- linear recurrence relation
- mapping
- modulus $|z|$
- monic polynomial
- monotonic decreasing function
- monotonic function
- monotonic increasing function
- monotonic sequence
- multiple root
- multiplicity
- nested function $f^k(x)$
- odd function
- one-to-one
- one-to-one correspondence
- onto
- order (recurrence relation)
- parabola
- period
- periodic function

- polynomial
- polynomial equation
- preimage
- purely imaginary number
- quadratic equation
- quadratic polynomial
- quartic polynomial
- quotient
- range
- real part $\operatorname{Re}(z)$
- recurrence relation
- recurrence sequence
- remainder
- restriction $f|_C$
- root
- sequence
- simple root
- solution
- strictly decreasing function
- strictly increasing function
- surjective
- unbounded function